

Absolute value equations



- 1 **a** $|x| = 9$ **b** $|5x - 2| = 3$
- 2 **a** 2 solutions **b** There are no solutions.
- 3 $b = 0$
- 4 Neither, because the equation has no solutions since the the left hand side of the equation can never be negative.
- 5 **a** $x = \pm 8$ **b** $x = \pm 10$ **c** $x = \pm \frac{4}{9}$ **d** $x = \pm 5$
e $x = \pm 40$ **f** $x = \pm 42$ **g** $x = \pm 6$ **h** $x = \pm \frac{2}{5}$
- 6 **a** $x = 4, -10$ **b** $x = 1, -11$ **c** $x = -8, 14$ **d** $x = -2, 4$
e $x = -8, 22$ **f** $x = -6, 14$
- 7 **a** $x = 2, -8$ **b** $x = 1, -5$ **c** $x = -2, -8$ **d** $x = 1, -9$
e $x = -\frac{1}{3}, 5$ **f** $x = -\frac{8}{5}, 2$ **g** $x = -4, \frac{2}{3}$ **h** $x = -1, -\frac{11}{4}$
- 8 **a** $x = \pm \frac{9}{2}$ **b** $x = \pm 7$ **c** $x = \pm \frac{1}{2}$ **d** $x = 5, -1$
e $x = \pm 5$ **f** $x = -3, -5$